Industrial Engineering & Operations Research
College of Engineering

IEOR 142A Machine Learning and Data Analytics I (4 units)

Spring 2026

Description:
This course applies foundational concepts and techniques in machine learning and data analysis to a diverse set of examples using real datasets from domains such as e-commerce, healthcare, social media, sports, the Internet, and more. Through these examples, exercises in Python, and a comprehensive team project, students will gain experience understanding and applying techniques such as linear regression, logistic regression, classification and regression trees, cross-validation, random forests, boosting, text mining, data visualization, time series modeling, clustering, principal component analysis, regularization, neural networks, and large-scale learning.

Note: *Students cannot receive credit for both IEOR 142A Introduction to Machine Learning and Data Analytics and IEOR 242A Machine Learning and Data Analytics.*

Instructor:
Prof. Phillip Kerger, kerger@berkeley.edu
Office hours: Mondays 1:15-2:30pm, Etcheverry Hall 4117,
or by appointment. You can find my calendar here (accessible with your Berkeley login) to find a time.

Teaching Assistants & office hours:

Chenran Weng (chenrann@berkeley.edu) OH: **Thursday 11-12 Etch 4176C.**

Jackson Gao (xgao@berkeley.edu) OH: **Tuesday 10-11, Etch 4176B.**

Aoxi Li (aoxi@berkeley.edu) OH: **Tue 11-12 at Etch 4176C.**

Reader office hours:

- Kenny Wongchamcharoen (pattaraphon.kenny@berkeley.edu)
and Sanchit Ram Arvind (sanchitram@berkeley.edu) : **Thursday 1-2pm, Etch 4176C**
- Constantin Ertel (constantinertel@berkeley.edu) : **Wednesday 11am-12pm, Etch 4176B**

Lecture:
MWF 12-1pm, Physics 4.

Discussion Sections:

F 8-9am in Social Sciences 166
F 3-4pm in Etcheverry 3108
F 4-5pm in Etcheverry 3108

Exams:
Midterm exam 1: February 27, in person in class.
Midterm exam 2: April 3, in person in class.
Final Exam: May 13th, 3-6pm (location TBD), in person.

Detailed information (content, policies, etc) and practice problems will be given at a time closer to the exam.

Prerequisites:

At least one course in **probability/statistics** (at the level of IEOR 141) and one course in **optimization** (at the level of IEOR 162); and thus "transitive" prereqs of **calculus 1-3 & linear algebra**. These are not strictly enforced, but these topics will be treated as known. Programming experience and literacy is expected, and will be required for implementations in the homework assignments.

Grading:
There will be regular homework assignments, two midterms, a final exam, and a team project. The grading rubric is flexible, and will be adjusted within the following stated bounds to maximize the grade of each student individually:

1. Homework: 15%
2. Midterm 1: 15% to 25% (in benefit of each individual student)
3. Midterm 2: 15% to 25% (in benefit of each individual student)
4. Project: 10%
5. Final exam: 25% to 45% (in benefit of each individual student)

For example, if your final exam score is higher than both midterm scores it would count as 45% of your grade, and each midterm would only count 15%. Another student might have 15% weight on midterm 1, 25% on midterm 2, and 35% on the final exam if that possible weighting yields the highest grade for that student.

Assignments:
There will be homework assignments assigned periodically during the semester which **can be submitted in groups of 2** to encourage you to work together (and help each other instead of going to ChatGPT). Please either work through the problems together, or each attempt all problems fully and then discuss. Please *do not* split up the homework and each do half of the problems or alternate doing the homeworks; that will just set you up for failure on the exams.

All assignments will be turned in using Gradescope. **No late work will be accepted**, (except potentially for documented DSP-related accommodations) however each exam will "clobber" the HW scores since the previous exam, so if you miss an assignment that grade will be replaced by your grade on the next exam.

Readings / Textbooks::
The main textbook for this course is:

- *An Introduction to Statistical Learning: with Applications in Python* by Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani, and Jonathan Taylor, Springer, 2023. A PDF version of this textbook is available for free.

All readings are recommended (not required) and will complement the lecture material. Most of the time, the readings will be in a different style and will cover more/different material than we will be able to cover in lecture. That is, the material covered in class is **not** a strict subset of the corresponding chapters in the textbook. Instead, the Venn diagram depicting the intersection of material covered in class and material in the textbook is substantial. Depending on your learning style, you may find it helpful to complete the readings either before or after the corresponding lecture.

Some other supplementary texts are:

- *The Analytics Edge* by Dimitris Bertsimas, Allison K. O'Hair and William R. Pulleyblank, Dynamic Ideas LLC, 2016.
- *The Elements of Statistical Learning: Data Mining, Inference, and Prediction* by Trevor Hastie, Robert Tibshirani and Jerome Friedman, Springer, 2009. A PDF version of this textbook is available at https://statweb.stanford.edu/~tibs/ElemStatLearn/. This is an advanced textbook and goes far beyond the material that we will be covering, but this could be a valuable resource for students with a strong mathematical background.
- *Python for Data Analysis, 3E* by Wes McKinney, O'Reilly Media, 2022. An online version of this book is available at https://wesmckinney.com/book/

Software:
The course will primarily use Python, mainly during the discussion sections/labs. This is *not* a programming course, rather software is a tool for us to work with data, build models, and synthesize our results. You are allowed to use any language/software that you want for the homework assignments, midterm, and final project. However, we recommend Python as it is the language that will be taught in the discussion labs and is the only language that is officially supported by the course staff. If and when you run into a programming issue, you are not only allowed to but encouraged to make use of the internet for fixing your code. There may also be tasks you are asked to do that are *not* covered in the course (e.g. making a particular type of figure), for which you should independently figure out how to accomplish them by making use of documentation, StackOverflow, etc.

Course Objectives:

1. To expose students to a variety of statistical learning methods, all of which are relevant and useful in a wide range of disciplines and applications.
2. To carefully present the statistical and computational assumptions, trade-offs, and intuition underlying each method discussed so that students will be trained to determine which techniques are most appropriate for a given problem.
3. Through a series of real-world examples, students will learn to identify opportunities to leverage the capabilities of data analytics and will see how data analytics can provide a competitive edge.
4. To train students in how to actually apply each method that is discussed in class, through a series of labs and programming exercises.
5. For students to gain some project-based practical data science experience, which involves identifying a relevant problem to be solved or question to be answered, gathering and cleaning data, and applying analytical techniques.
6. To introduce students to advanced topics that are important to the successful application of machine learning methods in practice, include how methods for prediction are integrated with optimization models and modern optimization techniques for large-scale learning problems.

Course Website and Communication:

Lecture materials, homework assignments, and all other course materials will be posted on the **bCourses** site. Additionally, we will use an **Ed Discussion** forum as the **main electronic communication method** for the course, including announcements. If you have any questions regarding the course, please post them on Ed Discussion rather than emailing the course staff. If you have a question or concern that is private in nature (i.e., something you would normally send as an email to the course staff), please use a private post on Ed Discussion so that only the course instructor and GSIs can see your message. You are encouraged to use public posts in situations where other students may benefit from the discussion. **Note that in "anonymous" posts on Ed, student names are visible to the course instructor.** In circumstances where your message should be kept confidential from the GSIs, please email the course instructor and begin the subject line with "[IEOR 142 Confidential]". In summary, you should observe the following priority list for course related communications:

1. Make a public post on Ed Discussion
2. Make a private post on Ed Discussion that only the course instructor and GSIs can see
3. In rare circumstances, send an email to the course instructor using "[IEOR 142 Confidential]" to start the subject line.

We ask that you also please observe the following etiquette on Ed Discussion:

1. **Do not post answers:** Please do not post any answers or your current results on Ed Discussion. Instead, you should explain the key points of your question in a way that allows other students to figure out the essence of the problem on their own, or guides them in the right direction. Post problem spoilers after the due date. If you think that your post might give out too much information about the problem solution, then make it private and let the course staff know.
2. **No pre-grading:** We will not answer any questions of the form "Is this the correct way to solve Homework X, Problem Y?"
3. **Aim for public posts:** Other students may have the same question, so please try to make your posts public.
4. **Formatting:** Please try to format code and mathematical equations using the appropriate tools. Here is a helpful Quick Start Guide.
5. **Broad Questions:** Broad conceptual questions that would require lengthy extensive answers might not be answered by course staff on Ed; office hours is the better option for these.
6. **Discussion and collaboration:** We highly encourage you to answer or comment on your fellow students' posts if you know the answer or would like to discuss.

Academic Ethics and Conduct:
You are expected to adhere to the UC Berkeley Code of Student Conduct at all times. In particular, please give credit to outside sources that you find helpful in completing the assignments. These include your peers or other people who you discuss your work with, other textbooks, material from other courses, etc. (There is no need to cite the course textbooks, slides, or other materials distributed on bCourses.)

Project:
There will be a final project that should be done in **teams of five students**. The project provides an opportunity for students to apply analytical methods to a problem in a domain of their choosing.

You will gather (and clean up) data relating to your chosen problem, and use the data analytics techniques discussed in class to solve/answer one or more substantive problems or questions. The project will give students some experience in the kind of work that a data scientist might perform in practice. The requirements for the project are outlined below:

- Each team must submit a one-page proposal that outlines a plan to apply analytical methods to a problem you identify using some of the concepts and tools discussed in the course. The proposal should include a description of: (1) the problem, (2) the data that you have or plan to collect to solve the problem, (3) which techniques you plan to use, and (4) the impact or overall goal of the project (if you could build a perfect model, what would it be able to do? are there any potential negative consequences to the development of such an ideal model?). The teaching staff will be available to answer questions, and will provide all students with electronic feedback.

- The final project submission will consist of a written report of at most **4 pages** (not including appendices) that describes the analysis, as well as a **5-minute** presentation (in powerpoint or pdf format) of your project. Project presentations will be either in person during the last week of classes or RRR week, or they will be recorded and posted on YouTube (private videos are admissible).
- The four-page report (not including appendices) that describes your analysis is due on the nominal day of the final exam.

Class Format and Participation:

Lectures will be held in person. Attendance is not part of the grade for this course but is highly encouraged; there is plenty of research showing attending live class leads to better learning outcomes. Recordings will be posted to accommodate when student's cannot show up, but these are not guaranteed.
Lecture slides (and any necessary code and data sets) will be posted on the bCourses if you wish to review them.

Discussion sections/labs, run by the GSIs, will be held in person and will be recorded as well. They will consist of interactive sessions that will cover additional examples of the methods presented in the lectures. You are encouraged to work interactively on your computer as your GSI runs live coding demos.

Academic Integrity:
Once again, you are expected to adhere to the UC Berkeley Code of Student Conduct at all times. Ethical conduct is of the utmost importance throughout your studies, career, and life and is much more important than grades. One of the most important values of an academic community is the balance between the free flow of ideas and the respect for the intellectual property of others. Again, please give credit to outside sources that you find helpful in completing your assignments for this class. Serious incidents of plagiarism, cheating, and other forms of academic misconduct will result in a penalty to your grade and will be reported to the Center for Student Conduct for further review and possible disciplinary action.

AI Tool Usage Policy:
The goal of this policy is to encourage responsible and ethical use of AI in educational settings, preparing students for both the benefits and challenges of these emerging technologies. It is imperative that students view AI tools as supplementary aids that enhance, but do not replace, their own critical thinking and problem-solving abilities.

1. I Tool Usage Allowed: Students can use AI for help with homework and coding assistance (especially coding, as AI-assisted coding is quickly becoming the industry standard).
2. For writing reports, summaries, interpretations etc, AI use is highly discouraged (we have no real way of policing this except in cases where it is obvious, but please don't have ChatGPT interpret your results for you when a question asks something along this lines of "what do you think about the results from [xyz]?").
3. Critical Evaluation: Students must critically evaluate and verify the accuracy of AI-generated content; you own the work that you turn in.
4. Prohibited in Exams: The exam will be in person and AI tools, in addition to other electronic devices, are not allowed in examinations.
5. Balanced Use Encouraged: Students should balance AI use with developing the ability to perform academically without it. They should learn to discern when the use of AI is beneficial and when it is not.

Tentative List of Topics (subject to some change):

- Introduction, motivating examples; basics of classification, function approximation and universal approximators; neural networks.
- Linear regression – model fitting, analyzing output, and model validation. In-depth application: predicting the quality of wine.
- Logistic regression, linear discriminant analysis, and ROC analysis. In-depth applications: predicting loan defaults and customer churn rates.
- Resampling techniques: Cross-validation and Bootstrapping
- Dimensionality reduction: PCA
- Classification and regression trees (CART); cross-validation.
- Classification trees - interpretability of ML methods
- Advanced tree-based methods: random forests and boosting.
- Regularization methods
- Unsupervised learning: Clustering
- If time: Introduction to generative models and LLMs

Time conflicts:

Students may enroll even if they have time conflicts. However, students will still be responsible for keeping up with the material, attending in-person when mandatory (usually exams only), and no special accommodations will be made for time conflicts. If you feel confident you can participate this way, confirm via email and I will assist you in enrolling. If you are uncertain about conflicts, please reach out to your major advisor, academic advisor, or faculty.

Inclusion:

We are committed to fostering a learning environment welcoming of all students. To do so, it is important that all participants of this class (instructors and students) support a diversity of perspectives and experiences and respect each other's identities and background (including race/ethnicity, nationality, gender identity, socioeconomic class, sexual orientation, language, religion, ability, etc.). If you feel like your performance in the class is being impacted by a lack of inclusion, please

contact the instructors, your academic advisor, or see the Academic Accommodations Hub for more resources.

Academic Accommodations:
UC Berkeley is committed to creating a learning environment that meets the needs of its diverse student body. If you anticipate or experience any barriers to learning in this course, please feel welcome to discuss your concerns with the instructors.

The Academic Accommodations Hub is the student and instructor portal for support resources and academic policies, including key information regarding academic accommodations, as well as links to mental health resources, resources for survivors of sexual violence and sexual harassment, and more.

If you have a disability, or think you may have a disability, you can work with the Disabled Students' Program (DSP) to request an official accommodation. The Disabled Students' Program (DSP) is the campus office responsible for authorizing disability-related academic accommodations, in cooperation with the students themselves and their instructors. You can find more information about DSP, including contact information and the application process here. If you have already been approved for accommodations through DSP, please contact the instructors so we can develop an implementation plan together.
Students who need academic accommodations or have questions about their accommodations should contact DSP, located at 260 César Chávez Student Center. Students may call 642-0518 (voice), 642-6376 (TTY), or e-mail dsp@berkelely.edu